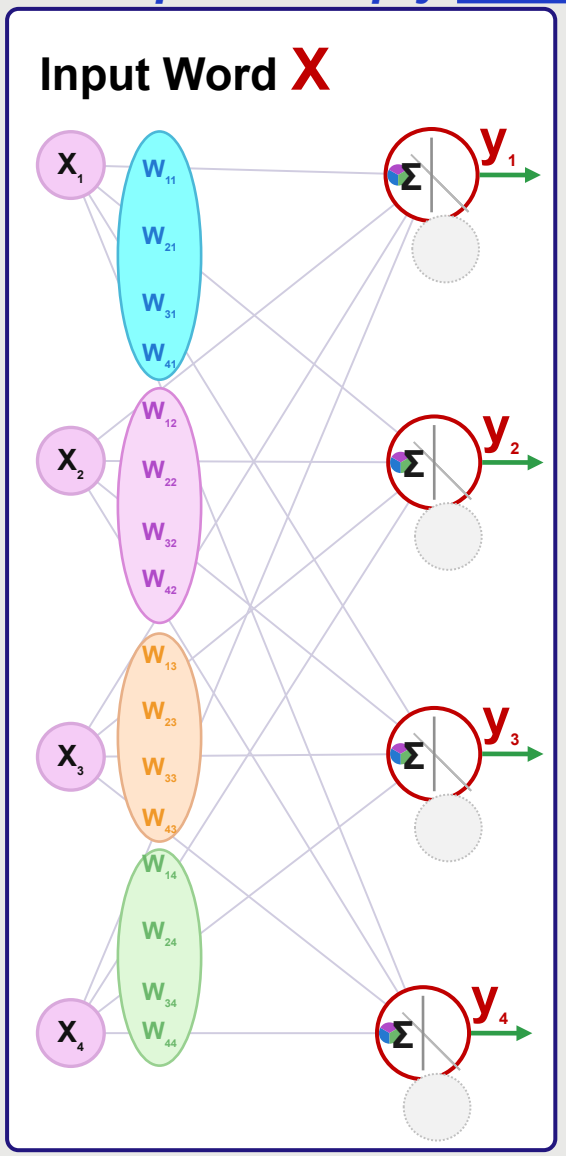
Using FF NN for Multiplication of ONE Vector by a Matrix

Example: Multiply Vector of Current Word X by W_Q (Query) {Remember: W_Q is Squared Matrix}



$$Y^T = \begin{bmatrix} y_1 & y_2 & y_3 & y_4 \end{bmatrix} = Sigmoid$$

$$Y^T = \begin{bmatrix} y_1 & y_2 & y_3 & y_4 \end{bmatrix} = Linear$$

IF No Non-linearity

$$\left(\begin{bmatrix} x_1 & x_2 & x_3 & x_4 \end{bmatrix} X \begin{bmatrix} W_{11} & W_{21} & W_{31} & W_{41} \\ W_{12} & W_{22} & W_{32} & W_{42} \\ W_{13} & W_{23} & W_{33} & W_{43} \\ W_{14} & W_{24} & W_{34} & W_{44} \end{bmatrix} \right) + \begin{bmatrix} b_1 & b_2 & b_3 & b_4 \end{bmatrix}$$

IF No Bias

$$\left(\begin{bmatrix} x_1 & x_2 & x_3 & x_4 \end{bmatrix} X \begin{bmatrix} W_{11} & W_{21} & W_{31} & W_{41} \\ W_{12} & W_{22} & W_{32} & W_{42} \\ W_{13} & W_{23} & W_{33} & W_{43} \\ W_{14} & W_{24} & W_{34} & W_{44} \end{bmatrix} \right) + \begin{bmatrix} 0 & 0 & 0 & 0 \end{bmatrix}$$